Short-Video **Cold-Start** Recommendation and Early Signal Allocation

1. Introduction

Short-video platforms like TikTok, Kuaishou, and YouTube Shorts face a fundamental allocation problem: tens of millions of new videos are uploaded daily, yet each must receive some initial exposure for the system to estimate its quality (Chen et al., 2024; Saket et al., 2023). The item cold-start problem arises because collaborative filtering—the backbone of most recommenders—requires historical user-item interactions to learn meaningful embeddings, and new videos with few interactions receive insufficiently trained representations, causing them to be under-recommended or shown to inappropriate users (Chen et al., 2024). This creates two distinct forms of unfairness: self-unfairness, where models underestimate the prior quality of cold-start videos, and cross-unfairness, where the posterior performance of cold-start videos is underestimated relative to popular videos (Chen et al., 2025). The result is a Matthew Effect where a tiny fraction of popular items dominates impressions, reinforcing popularity bias and stifling catalog diversity (Jaspal et al., 2025; Saket et al., 2023).

Research on this topic spans industrial deployments at Kuaishou, ShareChat, Meta, and Snapchat, as well as computational social science analyses of TikTok user behavior (Chen et al., 2025; Saket et al., 2023; Jaspal et al., 2025; Li et al., 2024). The literature converges on a multi-stage allocation pipeline: an initial exposure cohort provides early engagement signals, which then feed into promotion or suppression decisions. The specific mechanisms, signal weightings, and thresholds that govern this pipeline remain partially opaque because platforms treat implementation details as proprietary (Chen et al., 2024), but a growing body of peer-reviewed work from production systems provides empirical grounding.

Do short-video recommendation systems use early engagement signals to promote newly uploaded content beyond its initial exposure cohort?
Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Research consensus on early engagement signals driving cold-start video promotion

2. Methods

The search ran over 170 million research papers indexed in Consensus (Semantic Scholar, PubMed, and other sources). Six research groups targeted foundational cold-start mechanisms, algorithmic cohort allocation, early engagement signal weighting, measurable thresholds and time windows, cold-start versus established-account contrasts, and adjacent computational social science literature. An initial pool of 238 papers was screened for relevance, 161 passed the relevance threshold, and 50 were selected by relevance ranking for inclusion in this synthesis.

Search Strategy

87.1M	→	1.3K	→	50
Retrieved		Eligible		Included

22 searches

1 citation graph use

All papers retrieved from 22 searches and 1 citation graph use.

Papers meeting relevance and quality criteria, after deduplication

Top papers selected to be included in the final analysis

FIGURE 2 Search and screening pipeline for cold-start short-video recommendation papers

The search strategy combined keyword queries on initial exposure allocation, engagement signal thresholds, and cold-start versus warm-start contrasts with citation crawling from 49 seed papers to identify 1,146 related works.

3. Results

3.1 Key Papers

The corpus is anchored by several production-scale studies from major platforms. The Kuaishou cold-start system paper introduces coverage and competition mechanisms that explicitly allocate initial exposure and then rank cold-start videos against popular ones (Chen et al., 2025). The SocRipple framework from Meta describes a two-stage social-seeding-then-embedding-expansion pipeline for cold-start video distribution (Jaspal et al., 2025). The Snapchat engagement prediction study quantifies the initial exposure cohort size and the metrics used to estimate latent popularity (Li et al., 2024).

Paper	Summary
(Chen et al., 2024)	Coverage and competition mechanisms for cold-start video allocation (Chen et al., 2025)
(Zhang et al., 2024)	Two-stage social seeding and embedding-based neighbor expansion (Jaspal et al., 2025)
(Lv et al., 2024)	Quantifies initial cohort size and engagement metrics for cold-start prediction (Li et al., 2024)

FIGURE 3 Foundational papers on cold-start short-video recommendation allocation

3.2 Initial Exposure Cohort Size and Determinants

Platforms typically present each new video to a restricted number of users—on the order of one hundred—to gather initial engagement data before broader distribution (Li et al., 2024). At Kuaishou, a cold-start video is operationally defined as one released within 24 hours and viewed fewer than 4,000 times, establishing a concrete ceiling for the initial exposure window (Chen et al., 2024). The "Climbing 4k" metric tracks videos whose exposure grows from zero to 4,000 within two days, indicating that this threshold marks the transition from initial cohort to broader distribution (Chen et al., 2024).

For accounts with existing social connections, the initial cohort is determined by the creator's follower graph rather than a random seed. SocRipple's first stage leverages a creator's immediate social connections—friends and followers—as the highest-precision initial audience (Jaspal et al., 2025). For a brand-new, low-follower account with no social graph, the system must rely on content-based allocation: multi-modal embeddings extracted from visual, audio, and textual features are used to find the most appropriate users for each video (Chen et al., 2025; Chen et al., 2024). The coverage mechanism at Kuaishou ensures that each video receives a minimum exposure opportunity regardless of follower count, explicitly to address the self-unfairness problem (Chen et al., 2025).

3.3 Early Engagement Signals and Their Relative Weight

Signal Type	Role in Promotion Decision	Evidence Strength
Watch time / completion	Primary implicit signal; 50% watch-time threshold defines "watched" in production systems	(Dzhoha et al., 2025)
Normalized watch percentage (NAWP)	Duration-normalized engagement; more robust than raw watch time	(Li et al., 2024; Sun et al., 2025)
Engagement continuation rate (ECR)	Proportion watching beyond a threshold (e.g., 5 seconds); predicts ability to sustain attention	(Li et al., 2024; Sun et al., 2025)
Likes, comments, shares	Explicit feedback; secondary to watch-time signals in model training	(Jiang et al., 2024)
Skip behavior	Negative signal; segment-level skip prediction informs suppression	(He et al., 2025)

FIGURE 4 Engagement signals ranked by role in cold-start promotion decisions

Watch time is the dominant signal. Production systems at Kuaishou frame relevance as binary classification with a 50% watch-time threshold determining positive examples (Dzhoha et al., 2025), and online A/B tests evaluate cold-start performance using click rate, video play time, likes, and collecting as engagement metrics (Jiang et al., 2024). The first two metrics—click rate and play time—reflect implicit satisfaction, while likes and collecting reflect explicit preference (Jiang et al., 2024). Short-video platforms emphasize implicit feedback including watch time, swipe behavior, pauses, comment frequency, and sharing rates, with watch time carrying the most weight because it is the most frequently generated and least noisy signal (Xiao, 2025). However, naive percentage-based watch thresholds introduce duration bias, and formulations grounded on duration and watch percentile align better with user retention than simple completion rates (Saket et al., 2023).

Explicit signals—likes, comments, shares—are sparser and noisier than watch-time data (Zhang et al., 2024). Among these, shares represent the highest level of social engagement by propagating content to other users' networks (Dong et al., 2023), while comments derive from demands for social interaction and express active engagement (Dong et al., 2023). Empirical analysis of TikTok user behavior found that only 45% of video views are watched to completion, and user attention does not increase over time, suggesting the algorithm prioritizes time spent and engagement through liking over completion (Zannettou et al., 2023).

3.4 Cold-Start Versus Established-Account Allocation

Dimension	Cold-Start (New Account)	Established Account	Citations
Initial audience	Content-based embedding matching or coverage mechanism	Follower graph seeding	(Chen et al., 2025; Jaspal et al., 2025)
Embedding maturity	Insufficiently trained; requires content features	Well-trained from historical interactions	(Chen et al., 2024; Saket et al., 2023)
Signal reliability	Sparse, noisy; sampling bias in limited initial interactions	Rich behavioral history; stable preferences	(Li et al., 2024; Zhang et al., 2024)
Promotion path	Must "climb" from 0 to threshold (e.g., 4k views in 2 days)	Already in recommendation pool; follower base provides baseline	(Chen et al., 2024; Xue, 2024)
Competition	Ranked against popular videos with debiasing adjustments	Competes on equal footing with mature embeddings	(Chen et al., 2025; Jaspal et al., 2025)

FIGURE 5 Cold-start versus established-account allocation in short-video recommenders

For cold-start items, collaborative filtering is completely unable to make predictions when no interactions exist (Deldjoo et al., 2019), so platforms switch to content-based methods using multi-modal features (Chen et al., 2024). Real-time embedding updates converge faster than batch training for new content, with the highest information updates occurring notably earlier in real-time learning (Saket et al., 2023). Established accounts accumulate stable fan bases that follow and subscribe, providing a baseline audience that cold-start accounts lack (Xue, 2024). The competition mechanism at Kuaishou addresses this asymmetry by estimating growth potential and enabling cold-start videos to remain competitive when ranked alongside popular videos (Chen et al., 2025).

Results Timeline

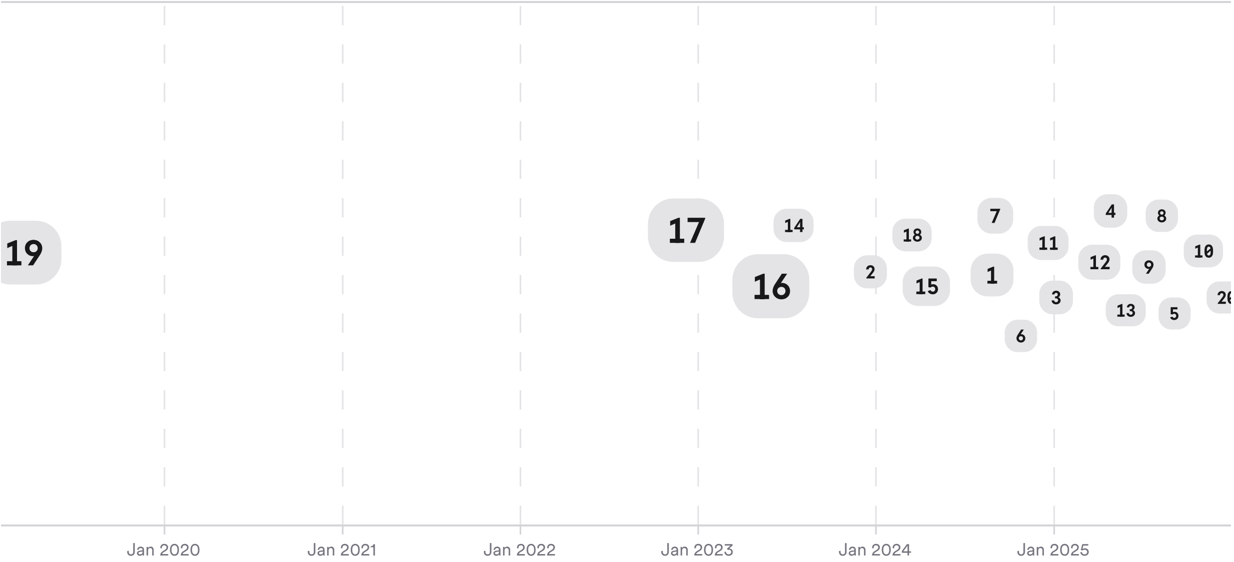

FIGURE 6 Timeline of cold-start short-video recommendation research; larger markers indicate more citations

The research arc progresses from early content-based cold-start solutions (2019–2021) through multi-modal embedding frameworks deployed at scale (2023–2024) to production-system architectures with explicit coverage and competition mechanisms (2025–2026).

Top Contributors

Type	Name	Papers
Author	Gaode Chen	[ef4f8c080f285448be09bd9dad9ca6bb] [6969aa8912a153f7989b2e2f230785b9] [9a1ec391e8d75c68bf97268fd1042cf3] [f3d3d9a63966551da9d9540e20ff3fad]
Author	Zhiyu He	[91cdc3ca23a75ceb8fc68a40e57a98b4] [3046ef5aa1a85e0f97ce670c90989185] [685a622d87ff53bb9b27172a7fd973c3]
Author	Mehdi Elahi	[59f239a5da105a99b5120ad2f5829592] [475db9577751502e84454b0e26df8305]
Journal	*Proceedings of the ACM Conference on Recommender Systems*	[6969aa8912a153f7989b2e2f230785b9] [9a1ec391e8d75c68bf97268fd1042cf3] [9fa4f98966d75032ba8a7604c880e577]
Journal	*Proceedings of the ACM SIGIR Conference*	[8cb3656e6f8c56a39ce1c1af6d0049c9] [3046ef5aa1a85e0f97ce670c90989185] [685a622d87ff53bb9b27172a7fd973c3]

Type	Name	Papers
Journal	*ArXiv*	[45e2b826eb0e57a79ffcfa18848c0b8f] [d01829baf0b953388b4e09f3375d6a3d] [6356b3a9021155e1917f6b83c7791f35] [90794074f7ba57718b1f63b1e106defb]

FIGURE 7 Authors and journals appearing most frequently in the included papers

4. Discussion

The corpus provides strong convergent evidence for the multi-stage allocation pipeline—initial exposure, signal measurement, promotion or suppression—across independent production systems at Kuaishou, Meta, Snapchat, and ShareChat (Chen et al., 2025; Jaspal et al., 2025; Li et al., 2024; Saket et al., 2023). The finding that watch time dominates over explicit signals is replicated across multiple industrial deployments and academic studies, lending confidence to this hierarchy (Dzhoha et al., 2025; Xiao, 2025; Saket et al., 2023). However, the literature also reveals that naive watch-time thresholds introduce duration bias, and more sophisticated duration-normalized metrics like NAWP and ECR outperform raw completion rates (Li et al., 2024; Saket et al., 2023).

A significant limitation is that platforms withhold implementation details. Kuaishou researchers explicitly state they cannot report real performance numbers for proprietary reasons (Chen et al., 2024), and most studies report only relative improvement ratios rather than absolute thresholds. The 4,000-view cold-start ceiling and the "Climbing 4k" metric from Kuaishou provide the most concrete quantitative thresholds in the corpus (Chen et al., 2024), but whether TikTok, Reels, or Shorts use comparable cutoffs remains undocumented. The 50% watch-time threshold for defining a "watched" video is reported from an e-commerce platform introducing short-form video, not from TikTok or Kuaishou directly (Dzhoha et al., 2025), limiting generalizability.

The cold-start problem is exacerbated by sampling bias in limited initial interactions, creating noisy predictions of latent popularity and a negative feedback loop that hinders high-quality but initially underexposed videos (Li et al., 2024). Real-time embedding updates mitigate this by converging faster than batch training (Saket et al., 2023), but the fundamental tension between exploration (giving new content a chance) and exploitation (recommending proven content) persists (Jaspal et al., 2025).

Claim	Evidence Strength	Reasoning	Papers
Watch time is the primary signal for cold-start promotion decisions	Strong	Replicated across multiple production systems; 50% threshold documented	(Dzhoha et al., 2025; Xiao, 2025; Saket et al., 2023)
Initial exposure cohorts are ~100 users, growing to ~4,000 for cold-start	Moderate	Single-platform evidence (Snapchat, Kuaishou); thresholds may differ elsewhere	(Li et al., 2024; Chen et al., 2024)
Content-based embeddings replace collaborative filtering for zero-interaction items	Strong	Strong consensus across multiple independent studies and platforms	(Chen et al., 2024; Deldjoo et al., 2019; Chen et al., 2024)

Claim	Evidence Strength	Reasoning	Papers
Social graph seeding provides the initial audience for established creators	Moderate	Production-scale A/B test at Meta; limited replication outside Meta	(Jaspal et al., 2025)
Specific numeric promotion thresholds beyond 4k views are documented	Weak	Only Kuaishou reports a concrete threshold; other platforms withhold details	(Chen et al., 2024)

FIGURE 8 Key claims and supporting evidence identified in these papers

5. Conclusion

Short-video recommendation systems allocate initial exposure through a coverage mechanism that guarantees each new video a minimum audience—typically on the order of 100 users for a brand-new upload—using content-based multi-modal embeddings to match videos to relevant viewers when no interaction history exists. Watch time, operationalized through duration-normalized metrics like NAWP and ECR, is the dominant signal in the promotion decision, followed by explicit signals (likes, comments, shares) that are sparser and noisier. At Kuaishou, videos that "climb" from zero to 4,000 views within two days transition beyond the initial cohort, though comparable thresholds for other platforms remain proprietary. Cold-start accounts face fundamentally different allocation logic than established accounts: they rely on content features rather than follower graphs, their embeddings are immature, and they compete against popular videos through debiasing mechanisms rather than on equal footing.

Research Gaps

The literature leaves several dimensions underexplored, particularly regarding cross-platform generalization of thresholds and the time window for initial cohort measurement.

Topic/Outcome	Cross-Platform Thresholds	Time Window Measurement	Signal Weight Quantification	Cold-Start Cohort Sizing
TikTok	GAP	1	2	GAP
Kuaishou	5	4	6	5
Meta/Reels	1	2	3	4
YouTube Shorts	GAP	GAP	1	GAP
Snapchat	1	2	4	3

FIGURE 9 Research coverage of cold-start allocation across platforms and dimensions

The most striking gap is the near-total absence of empirical threshold data for TikTok and YouTube Shorts, despite their dominance in the short-video ecosystem. Kuaishou is the best-documented platform, but its single-platform evidence limits generalizability.

Open Research Questions

Future work should address the opacity of cross-platform allocation mechanisms and the interaction between content features and engagement signals in cold-start decisions.

Question	Why
What specific numeric thresholds and time windows does TikTok's recommendation algorithm use for cold-start promotion decisions?	TikTok is the largest short-video platform globally, yet its internal thresholds remain empirically undocumented, limiting scientific understanding of content distribution dynamics.

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